When Background Noise Has Meaning

Environmental Sound Classification under Semantic Interference

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Abstract

Environmental recordings can contain two recognisable events, although a classifier must often return one label. This project asks how different audio representations fail under generic noise and a competing labelled event at controlled relative power. A handcrafted SVM, compact log-Mel CNN and frozen LAION-CLAP receive identical mixtures from all five official ESC-50 folds. White noise strongly disrupts the SVM, whereas CLAP retains much higher performance under background noise yet assigns 60–67% of its errors to a competing event at +5 dB. Training with ordinary noise gives limited transfer to event competition; local feedback on the same frozen checkpoint adds no confirmed primary benefit. Two further analyses refine this conclusion: about one-third of CLAP's original captures retain the target at rank two, and small subsets matched on measured acoustic proxies retain a CLAP disadvantage but inconclusive CNN contrasts. These are controlled behavioural comparisons, not causal isolation of semantic meaning. The contribution is a reproducible DSP experiment that identifies where robustness gains transfer, where decisions move, and which explanations the data cannot establish.

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Project materials. Project website, working code and run instructions, and code demonstration video.

1 Introduction

This study uses all 2,000 five-second clips in ESC-50, a public dataset of 50 environmental-sound classes assembled from Freesound recordings [4]. Target events and semantic distractors come from ESC-50; real backgrounds come from DEMAND [12], and white noise is generated programmatically. The following dog-bark and alarm example illustrates the mixing principle.

A dog bark mixed with white noise contains one designated event and an acoustically nonspecific interferer. A dog bark mixed with an alarm contains two plausible class identities. Both mixtures can have the same power ratio while producing different errors. This distinction matters when deploying classifiers beyond single-label benchmarks, and recent benchmark work on structured acoustic scenes motivates looking beyond Gaussian noise [11]. Spectrogram CNNs and audio–language pretraining offer different

representations [5, 7]; the reviewed methods do not by themselves establish how their decisions move toward a known competing class under the same controlled mixtures. This is the specific comparison undertaken here.

The main research question is: *How do handcrafted, supervised spectrogram and pretrained audio–language representations fail under generic noise versus a competing labelled event at controlled relative power?* Error destinations and the transfer of noise augmentation are subordinate tests of this question. The intended contribution is to explain measured failure patterns rather than identify a universally best model. In particular, CLAP's large external pretraining makes equal-data or equal-training-cost comparisons inappropriate.

The experiment holds mixtures fixed and uses gain controls, alternative power definitions, repeated CNN training and rank/acoustic diagnostics. The proposal's power convention remains a full sensitivity analysis. All original predictions, including negative findings, are retained. Scope is one benchmark, three fixed systems, four interference types and three levels.

2 Literature review

2.1 From course DSP to environmental-sound classification

The course's audio-feature material distinguishes temporal energy and zero-crossing measurements from spectral descriptors and MFCCs [1]. These quantities compress a waveform into interpretable measurements. Their advantage is that a failure can be related to a change in a descriptor. Their limitation is that averaging descriptors across time discards event order and can mix properties from multiple sources.

The STFT lecture separates window length, hop size and FFT size [2]. The window controls the time–frequency trade-off, the hop controls frame spacing, and the FFT size controls frequency-grid spacing. Increasing the FFT length by zero padding does not recover frequency detail absent from a short window. These distinctions determine what transient and spectral information reaches both the handcrafted and CNN systems.

Piczak's environmental-sound CNN learns local spectro-temporal patterns from segmented spectrograms and their deltas [5]. This motivates a learned time–frequency reference alongside the handcrafted baseline. Its strengths are locality and trainable features, while the small labelled dataset limits supervision. The official ESC-50 record reports 39.60% accuracy for the MFCC/ZCR SVM and 64.50% for Piczak's CNN [4, 5]. Its supervised-results table also lists 98.25% for a model pretrained with natural-language supervision [4]. That protocol differs from frozen zero-shot evaluation. These are context, not matched implementations; the present CNN retains the representation principle rather than reproducing its architecture.

Salamon and Bello show that augmentation can improve environmental-sound CNN classification, with effects that depend on the augmentation and class [6]. This makes augmentation a testable intervention rather than a universal robustness guarantee. Here the question is whether training against white noise transfers to a second labelled event, and whether event-mixture training produces different trade-offs. Tokozume et al.'s between-class (BC) learning mixes different-class sounds and trains the output to match their proportions [8]. Our semantic augmentation instead keeps the designated target label: it tests target retention under interference, not soft mixture-label prediction.

2.2 Audio–language models and structured interference

CLAP learns a shared audio–text embedding space and supports zero-shot classification by comparing audio embeddings with textual class descriptions [7]. It supplies a contemporary pretrained reference without training another ESC-50 classifier. Text scores also reveal whether a competing label gains prominence. External training data and prompt sensitivity complicate interpretation, so the checkpoint and core prompt remain fixed. The revised paper reports 88.0–91.0% ESC-50 accuracy across its selected models, using `This a sound of label.` and excluding overlapping evaluation samples (Table 4, Section 4.3 and Appendix H) [7]. Here `630k-audioset-best.pt` uses `the sound of {label}` on all 2,000 clips. The published rows are not identified as this exact checkpoint/protocol; a score difference cannot be assigned to prompt wording alone.

PAT refines both modalities through audio–text feedback, including prompt weighting and frame-level audio reweighting, and evaluates robustness to degraded audio [9]. The fixed prompt ensemble tests the text side with cached audio unchanged. A separate local-feedback extension ports the released inference arithmetic to the same frozen LAION checkpoint, retaining uniform prompts. Its per-frame projection and preset coefficients are project choices; this is a PAT-style adaptation, not an official or full PAT reproduction.

Mix-CLAP's noise-aware audio-language formulation motivates the joint interest in robustness and deployment cost [10]. It is contextual literature rather than a reproduced system. RSA-Bench further motivates realistic structured backgrounds, but evaluates speech perception and reasoning under simulated acoustic scenes [11]. Its task differs from ESC-50 classification, so it supports the research motivation without supplying a directly comparable baseline.

2.3 Gap, method choice and concepts clarified

These works establish useful representations and robustness methods, but do not by themselves answer this project's controlled comparison of generic corruption and a known competing class. Three complementary systems are therefore retained: interpretable DSP features, a task-trained spectrogram model and a frozen audio–language representation. Holding the input mixtures fixed makes their failure patterns comparable, while separate cost measurements preserve the practical deployment question. Public code is used where reimplementation would add nothing to the question: the official LAION-CLAP architecture and pretrained weights are used without fine-tuning through the project inference wrapper, and PAT supplies the released prompt list and the feedback arithmetic adapted here. Piczak's network is not reproduced, and the STFT, features and mixer are written for this project so that every processing step can be inspected.

Three conceptual distinctions guide the implementation. First, decibels of relative component power are not perceived dominance: the hearing material discusses frequency-dependent sensitivity [3]. Second, cosine similarity and logits are scores rather than calibrated event probabilities. Third, a smaller regression coefficient after adjustment is not a percentage of a mechanism explained, partly because logistic coefficients are non-collapsible [13]. These distinctions motivated explicit power sensitivity, exact rank readout and cautious interpretation of acoustic adjustment.

3 Methods

3.1 Data and evaluation splits

ESC-50 contains 2,000 five-second clips, 50 balanced classes and five official folds [4]. For test fold k, the next cyclic fold is validation and the remaining three support model selection. A fresh model is then fitted on the four non-test folds. Scaling and Mel normalisation use the corresponding training partition. Four source IDs cross official folds, affecting nine target clips; official assignments remain intact, but those sources cannot supply semantic distractors. The resulting evaluation is not perfectly source-independent. The dataset's partly class-dependent preprocessing is a further caveat [4].

3.2 Signal processing and classifiers

At 44.1 kHz, the custom STFT uses a periodic Hann window of 1,024 samples, hop 512 and FFT length 1,024:

$$X(m,k) = \sum_{n=0}^{1023} x[n+512m]w[n]e^{-j2\pi kn/1024}.$$

Only the final incomplete frame is zero-padded. Windowing and framing are checked against SciPy and an actual execution of the course's unchanged `MyStft.m`. After reconciling frame alignment, window and one-sided output, the relative complex error on one ESC clip is approximately 2.6×10^{-16}. This validates corresponding frames, not identical default boundary policies.

System	Representation	Fitting and scoring
SVM	Means/SDs of RMS, ZCR, centroid, spread, roll-off, flux, 20 MFCCs and deltas: 92 values	RBF kernel; training-only scaling; nine C, γ candidates selected by clean validation
CNN	64 log-Mel bands; three convolution blocks (16/32/64 channels); 256-dimensional embedding	36,258 parameters; Adam; at most 40 epochs, patience 8; clean validation chooses duration
CLAP	HTSAT-tiny/RoBERTa, non-fusion `630k-audioset-best.pt`; unit embeddings	Frozen; 48-kHz resampling; official int16 truncation; cosine scores for `the sound of {label}`

CLAP's official frontend repeat-pads the canonical five-second input to ten seconds, and one fixed template serves every class so that prompt wording is not an uncontrolled variable. MFCCs decorrelate log-Mel energies by a cosine transform; the CNN retains the time–frequency map instead. The core CNN seed is 5305; repetitions use 5306 and 5307. Score magnitudes and absolute embedding distances are not compared across architectures.

3.3 Controlled mixing and power sensitivity

Every target receives white noise, a segment from three DEMAND recordings (traffic, station, metro) [12], and one same- and one different-coarse-category ESC distractor. Semantic partners have a different fine label and source, and come from the held-out fold, so neither supervised model has seen the interfering recording. Pairing is seeded, uses zero offset and stays fixed across +10, +5 and 0 dB. The noise waveform or segment also stays fixed across levels.

Let M_x, M_d be activity masks using frames within 40 dB of each signal's peak frame energy, and let $P(z; M)$ be mean-square power on mask M; a five-second average would understate sparse events such as a door knock. Here L denotes SNR for noise backgrounds and SIR for labelled-event mixtures. Mixing uses

$$\alpha = \sqrt{\frac{P(x; M_x)}{10^{L/10} P(d; M)}}, \qquad y = g(x + \alpha d), \qquad g = \min\left(1, \frac{0.99}{\max|x + \alpha d|}\right).$$

The control is gx. The default *own* convention uses $M = M_d$; the proposal's *target* convention uses $M = M_x$. Sparse distractors caused very large gains under the latter, motivating the documented change and full two-protocol comparison. Ratios are numerically verified. There are 24,000 own and 23,991 target mixtures: three zero-energy pairs exclude nine target-protocol rows. At 0 dB the two events have equal power, so the single designated label is ambiguous and this level serves as a stress test.

3.4 Metrics, uncertainty and extension design

Accuracy and macro-F1 are averaged over official test folds; macro-F1 is reported because degraded systems can concentrate their predictions on a few classes. Capture rate is the fraction of *errors* assigned to the distractor; capture counts over *all trials* are also retained. The margin is target score minus distractor score. Representation displacement compares a mixture with its gain-matched control, standardising handcrafted features by training-fold scales or using within-model cosine distance.

Comparisons retain the same targets, with source-cluster resampling to expose dependence. Fold and seed SDs describe variation, not confidence intervals. The three-seed overlap replication and training-intervention families use their declared Holm corrections (12 and 156 tests respectively). Intervals for the rank and acoustic analyses are pointwise 95% intervals from 2,000 target-source bootstrap resamples, conditional on fitted models and observed partners.

For the extension, SVM/CNN training uses either white-noise or semantic augmentation drawn only from training partitions. CNN augmentation probability is 0.5, with +5/+10/+15 dB training views; clean validation and architecture remain fixed. All three seeds are retained. A separate 300-entry fixed prompt ensemble reuses CLAP audio embeddings. No test-label prompt search or CLAP fine-tuning is performed.

Local feedback fuses pooled and guided scores from 32 frozen-projected HTSAT frames (heuristic temperature/weights 64/0.2/0.05). Calibration searches 321 settings on non-test clean folds, excludes shared sources and freezes one fold rule across interference; it is not zero-shot. Four primary own/+5-dB tests share Holm correction.

4 Results

4.1 Core robustness and error destinations

Table 1: Clean: fold mean ± SD (%). Capture: pooled captures/errors at own/+5 dB, 2,000 targets per condition; CNN seed 5305. The denominator is errors, not all trials.

	Clean (%)		Capture (%; captures/errors)	
System	Accuracy	Macro-F1	Same category	Different category
SVM	48.90 ± 4.21	47.96 ± 4.07	18.53 (283/1527)	15.89 (246/1548)
CNN	67.90 ± 3.65	66.61 ± 3.91	31.26 (403/1289)	27.53 (362/1315)
CLAP	81.75 ± 2.58	79.49 ± 3.28	66.95 (632/944)	60.30 (612/1015)

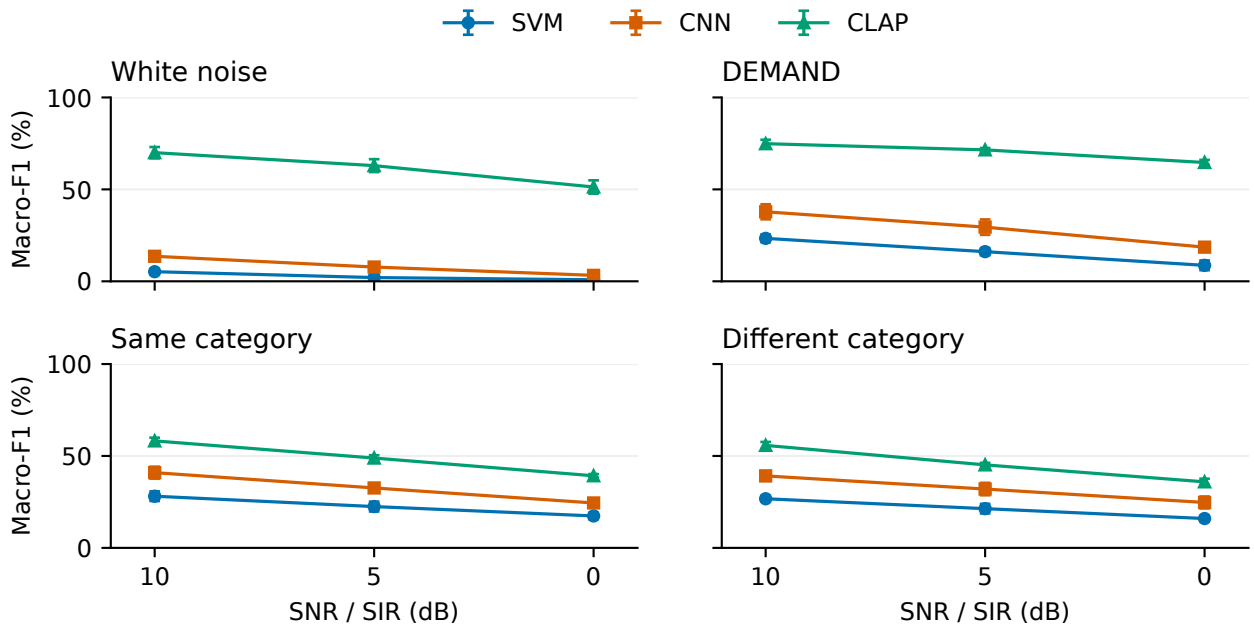


Figure 1: Own-activity robustness: mean macro-F1 with five-fold SD, not a confidence interval. Each condition/level has 2,000 targets. Lower relative level means stronger interference. Target-protocol curves are in the repository.

At +5 dB, SVM macro-F1 is 2.03% under white noise versus 22.47% under same-category interference. CLAP instead reaches 71.49% under DEMAND versus 48.86% under same-category interference. Equal nominal power does not give equal degradation; semantic mixtures add a known competing label.

Error destinations distinguish the systems (Table 1). The designated distractor receives 66.95% and 60.30% of CLAP's errors, against 18.53% and 15.89% for the SVM; uniform confusion over 49 wrong labels would give about 2%. For CLAP at own/+5 dB, the same-category median target–distractor cosine margin is 0.0873, with 38.05% negative, and median embedding displacement is 0.1905 for correct versus 0.4087 for incorrect trials; this is association, not causation. For the SVM, median standardised ZCR displacement is 3.16 under white noise against 0.25–0.29 under the other three conditions, consistent with broadband noise changing zero crossings and spectral-shape descriptors.

Figure 2 extends these comparisons across levels. Capture rises as the SIR falls for every system, from 56.25% to 71.45% of CLAP's same-category errors, while median margins move toward zero at 0 dB. The most frequent CLAP captures at +5 dB among pairs whose control was correct, selected from observed counts, are door knock to keyboard typing (10 of 11 pairs) and can opening to clock alarm (6 of 6). Figure 3 shows larger median displacement among misclassified mixtures in all 12 system/condition comparisons; individual distributions overlap. White noise displaces the SVM and CNN most, whereas CLAP's misclassified semantic mixtures move about as far as its misclassified white-noise mixtures (medians 0.41–0.43).

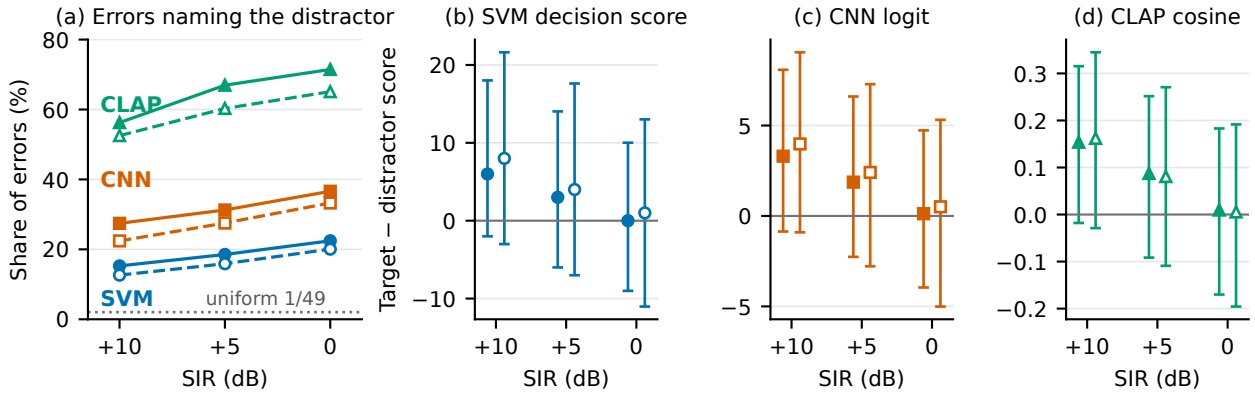


Figure 2: Own-activity semantic interference across levels. (a) Share of each system's errors that name the designated distractor; the dotted line is uniform confusion over 49 wrong labels. (b–d) Median target-minus-distractor score with interquartile range; scales are specific to each system. Filled/open markers: same/different category; panel (a) also uses solid/dashed lines. Each level uses 2,000 targets; margin bars show their score IQR, not a confidence interval.

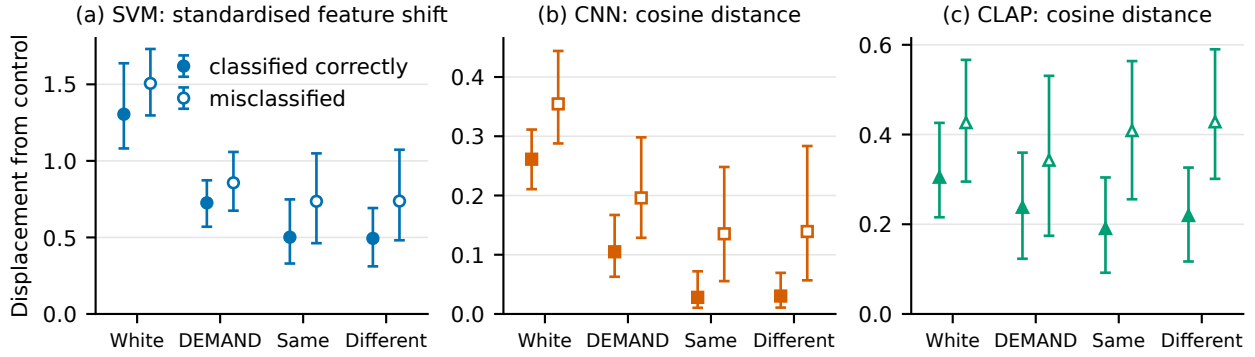


Figure 3: Displacement of each mixture from its gain-matched control at own/+5 dB: median and interquartile range for correctly classified (filled) and misclassified (open) mixtures. Each condition has 2,000 trials partitioned by outcome (correct groups: 108–1,479 trials; errors: 521–1,892). These are descriptive, outcome-conditioned distributions; scales differ across systems.

4.2 Overlap, training interventions and readout

The whole-dataset CNN ordering changes when attention is restricted to simultaneous events. On the original own-protocol high-overlap subsets (≥ 0.8), semantic-minus-matched-DEMAND accuracy differences are negative for all three CNN seeds. For seed 5305 at +5 dB, the differences are -3.28 points on 1,464 same-category targets and -2.94 on 1,465 different-category targets (both Holm $p = 0.0604$). Both additional seeds pass the declared correction; seed 5305 does not. This supports directional replication without treating every seed as a confirmed effect. Smaller acoustic-closeness subsets are examined separately below. Under target-activity power, +5 dB semantic macro-F1 is 0.47–2.35 percentage points lower than under own power, with the same system ordering. These are protocol comparisons with slightly different retained cohorts (1,998/1,999 same/different targets versus 2,000 each), not an isolated causal effect of the power mask.

Table 2: CNN augmentation: mean macro-F1 ± sample SD across three seeds. Own power; interference at +5 dB. Clean uses the original clips.

Training	Clean	White	DEMAND	Same	Different
Clean	65.78 ± 0.75	7.48 ± 0.42	30.92 ± 1.35	32.99 ± 0.34	31.41 ± 0.51
White noise	64.65 ± 0.61	43.95 ± 2.67	33.80 ± 2.19	33.46 ± 0.98	32.10 ± 1.05
Semantic	61.70 ± 2.40	9.91 ± 0.24	33.62 ± 1.55	36.05 ± 0.83	33.73 ± 0.86

White-noise augmentation changes CNN white-noise macro-F1 from 7.48% to 43.95%, but same-category performance only from 32.99% to 33.46%. Semantic training reaches 36.05% while clean performance falls

to 61.70%. Direct CNN semantic-over-white augmentation accuracy advantages do not pass the 156-test correction. SVM augmentation and all individual seeds are retained in the repository.

Fixed prompt ensembling raises CLAP clean accuracy from 81.75% to 92.90%. At own/+5 dB, same-category distractor captures change from 632 to 645 while other errors fall from 312 to 189. Thus a higher fraction of remaining errors can be captures even when overall accuracy improves. This text-side intervention precedes the separately tested local-feedback adaptation.

Table 3: CLAP readouts: clean and own/+5-dB same-category interference, each with 2,000 trials. Accuracy, macro-F1 and target top-two inclusion are percentages; captures and other errors are counts. Calibrated uses non-test clean labels.

Readout	Clean acc.	Same acc.	Same F1	Captures	Other errors	Target top-2
Single prompt	81.75	52.80	48.86	632	312	70.00
Fixed ensemble	92.90	58.30	55.50	645	189	78.20
Local: preset	92.90	58.25	55.46	644	191	76.90
Local: calibrated	92.85	58.20	55.47	645	191	76.10

The extraction reproduces both original readouts on all clean, mixed and control inputs. Relative to the fixed ensemble, preset alignment changes own/+5-dB accuracy by −0.05/+0.20 points (same/different); calibration changes it by −0.10/+0.75. No primary test passes Holm correction (smallest adjusted $p = 0.735$), and all pointwise source-cluster intervals, conditional on the selected rules, include zero. Fused scores are not cosines. Same-category target top-two inclusion falls from 78.20% to 76.90%/76.10%; calibrated clean accuracy is 92.85% versus 92.90%. This adaptation establishes no primary improvement; it does not test the published PAT model or weighted-prompt method.

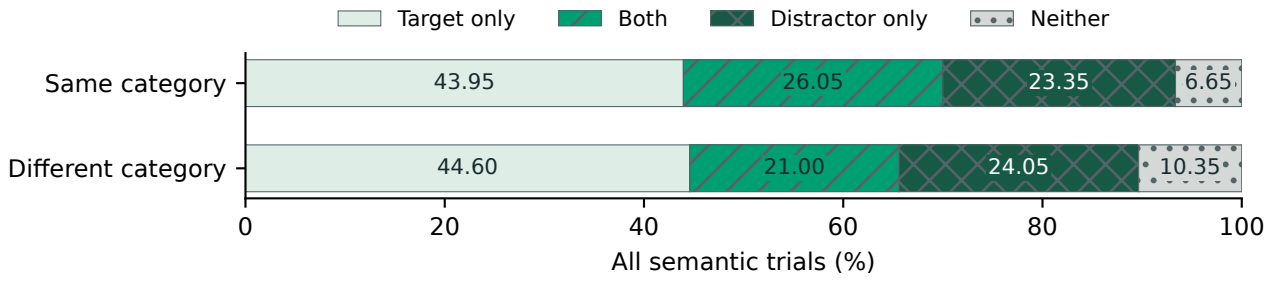


Figure 4: Single-template CLAP at own/+5 dB. Exact top-two membership of the two designated labels, among all 2,000 semantic trials per type. The four outcomes partition the trials; returned extra labels are not exhaustively annotated events.

Among original argmax captures, the target remains second in 236/632 (37.34%) same-category and 203/612 (33.17%) different-category cases. These denominators differ from the all-trial percentages in Figure 4. Gain-matched controls contain both labels in the top two in only 5.40% and 1.70%. Target-power results show similar partial recovery. Rank inclusion is not a calibrated detector or proof that the model perceptually heard both sources.

4.3 Spectral and temporal overlap

All 47,991 mixtures, including 12,000 DEMAND rows, receive spectral and temporal descriptors. Raw dB Mel correlation is not used for matching: unequal Mel-band widths make white noise appear anticorrelated with the target (median −0.728), which bandwidth normalisation removes (−0.008).

To compare the same target under semantic interference and DEMAND, the primary analysis retains pairs with differences at most 0.10 in target-active spectral cosine, 0.05 in the interference-dominated time–frequency fraction, and 0.10 in activity overlap. Selection uses acoustics only, identically for all three systems. These are acoustic proxies from the project frontend, not CLAP's internal frontend or a perceptual masking model.

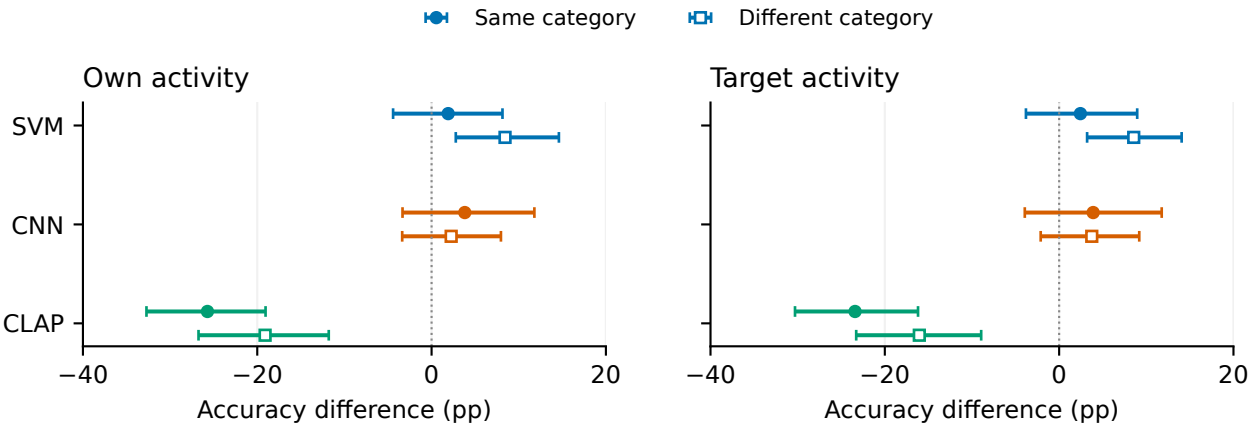


Figure 5: Semantic minus DEMAND accuracy at +5 dB, with pointwise 95% target-source cluster intervals (2,000 resamples). Own: 210/178 targets, 199/168 source clusters, 49/48 classes; target: 205/187 targets, 195/178 clusters, 49/47 classes (same/different). Filled circles: same category; open squares: different category.

Across both power conventions, CLAP remains about 16–26 accuracy percentage points below DEMAND, whereas all four CNN intervals include zero. Only about 9–10.5% of the benchmark remains. These selected populations differ from the larger high-overlap subsets; the new comparison is not a causal decomposition or a reversal on identical targets. Unmeasured acoustics and salience remain plausible explanations.

4.4 Deployment cost

Table 4: Median CPU query time (ms) on the same Apple M3, one numerical thread, batch one. Each pass contains 250 class/fold-balanced clips and three repetitions per model. Both model orders are retained.

System	First pass	Reversed order
SVM	15.00	11.29
CNN	3.07	2.44
CLAP	69.30	50.86

Total query cost includes frontend and scoring, excluding file access, model loading and one-off text preparation. Absolute times drift, but the CNN/SVM/CLAP ordering persists. CNN and CLAP have 36,258 and 158,334,169 parameters respectively. This is a local deployment comparison; external pretraining cost and possible data overlap are separate limitations. A separate identical-clip readout panel (750 queries per readout/order) gives ensemble 59.16/60.34 ms, preset alignment 61.74/63.01 ms and calibrated alignment 61.85/63.03 ms (forward/reverse medians). Compare within this new panel; the 125.89-s clean pass and 0.277-s calibration on cached scores are one-off preparation, excluded from query time.

5 Discussion

5.1 What the evidence answers

The three questions receive direct answers. First, the same nominal power degrades the systems differently: at +5 dB the SVM falls to 2.03% macro-F1 under white noise but keeps 22.47% under a same-category event, whereas CLAP keeps 71.49% under DEMAND but falls to 48.86% under a same-category event. Second, white-noise augmentation protects mainly against nonspecific noise: CNN macro-F1 rises from 7.48% to 43.95% under white noise but only from 32.99% to 33.46% under a same-category event. Third, when CLAP fails it usually predicts the interferer, in 66.95% and 60.30% of its errors.

These answers describe observed behaviour. The SVM's white-noise sensitivity can be traced to interpretable descriptors, as the course feature material anticipates [1], but displacement magnitude alone does not determine a decision; direction relative to class boundaries also matters.

Clean accuracy is 48.90% for our SVM and 67.90% for our CNN, respectively 9.30 and 3.40 points above the historical 39.60%/64.50% records [4, 5]. Feature conventions, architecture and training differ, so these are contextual comparisons rather than evidence of a controlled improvement. ESC-50's 81.30% crowdsourced human accuracy is also a different evaluation, not a matched human-parity test [4].

The original high-overlap comparison shows how a whole-dataset average can hide simultaneous-source difficulty. The later acoustic-closeness analysis narrows that claim: a negative CNN contrast is not established within its much smaller selected population. Keeping both analyses prevents convenient selection of a single favourable view.

5.2 How the interventions relate to prior work

The strong white-noise benefit and limited semantic transfer extend the augmentation question raised by Salamon and Bello [6]. Their reported augmentation benefits do not imply invariance to every corruption. BC learning trains a proportional two-class target [8]; our retained-target objective asks a different robustness question. Semantic training here exchanges some clean performance for semantic point-estimate gains. The corrected direct comparison remains inconclusive, so it would be misleading to recommend semantic training as uniformly superior.

Our CLAP accuracy changes from 81.75% to 92.90% with the fixed prompt ensemble on the same 2,000 clips. This is direct evidence of readout sensitivity within our protocol; it is not a superiority claim over the paper's 88.0–91.0% overlap-filtered results [7]. The top-two analysis adds a distinction between loss of the target's first-place decision and loss from the returned pair. Approximately one-third of original captures retain the target at rank two; many do not. The local-feedback extension tests released PAT-style arithmetic on the same checkpoint [9], but finds no confirmed primary improvement. Its uniform prompts and per-frame projection differ from the paper; this negative result concerns the adaptation, not a refutation of PAT. Neither changing prompts nor returning two labels retrospectively changes the original single-label evaluation.

A zero-shot readout scores the audio embedding against every class description [7], so a second event that matches one of those descriptions is a competing answer, not noise. This is consistent with the capture pattern but does not prove it. The remaining CLAP disadvantage in acoustically close subsets motivates further investigation of competing concepts. It does not prove a semantic mechanism: the measured descriptors are incomplete, selected target sets are small and shared interferers introduce dependence. RSA-Bench motivates attention to structured backgrounds [11], but differences in task, model families and scenes prevent direct numerical agreement claims. Likewise, the cost study addresses a concern relevant to noise-aware audio-language work [10] without reproducing that model.

5.3 Limitations and how to resolve them

There is one semantic partner per category type and only three DEMAND environments. Multiple pairing seeds and independent backgrounds would measure sampling sensitivity. Official source reuse and unknown CLAP pretraining overlap limit generalisation. Fixed alignment and three power levels cannot identify precise perceptual switching thresholds. A future within-pair onset/SIR sweep would address that limitation. Stronger mechanism tests would require acoustic controls whose recognisability is independently validated. At 0 dB, exhaustive multi-event annotation would better match the ambiguity of a scene. These are future directions, outside the completed experiment. The own-activity convention followed mixing diagnostics; top-two readout and acoustic-closeness thresholds followed inspection of model results. These analyses are exploratory. Starting again, these choices and several pairing seeds would be fixed before evaluation.

6 Conclusion

Equal-power mixtures produce representation-specific failures. White noise strongly disrupts the SVM, while CLAP errors frequently name a competing event. CNN white-noise augmentation offers limited transfer to event competition. Top-two readout retains some captured targets; prespecified local feedback establishes no primary gain. Gain controls and both power conventions make comparisons inspectable.

7 Implementation, reproduction and attribution

7.1 Real code demonstration

The accompanying execution video demonstrates one recorded example: can opening with a weaker clock alarm at +5 dB under own-activity power. The demonstration code loads the source recordings, reconstructs the mixture, checks achieved SIR and peak level, and executes the saved fold-matched SVM/CNN and single-template frozen CLAP. It displays fresh predictions and class ranks, then permits playback of the gain-matched control and mixture. Saved full-experiment outputs are used only afterwards to check agreement. The author also confirmed that representative website controls and mixtures were recognisable without obvious distortion; this is not a listening-accuracy study.

This failure case is illustrative, not representative: all controls predict can opening, while the mixture makes the CNN and CLAP select clock alarm and the SVM chirping birds. A run guide documents setup, data, commands and verification outputs.

7.2 Implementation contribution and learning

Project-authored components cover framing/STFT, feature extraction, compact CNN integration, power-controlled mixing, reproducible manifests, paired evaluation and diagnostics. External components include numerical/learning libraries, ESC-50/DEMAND and the frozen LAION-CLAP model. The PAT prompt list and released feedback arithmetic supply separately named text and local-alignment extensions. Attribution and exact versions are retained in the repository.

Implementation clarified why window length differs from FFT spacing, why sparse events require an explicit power mask, and why fold-local scaling prevents information leakage. STFT agreement, ratio checks, matched controls and independently recomputed statistics make these assumptions inspectable rather than dependent on library defaults.

7.3 Reproduction and attribution

The project website provides this report, its figures, listening examples, the video and working code. The repository preserves both power protocols, all seeds, score definitions, figure data and failure analyses. Checkpoint and dataset hashes identify evaluated inputs. Full datasets and large model weights are obtained through the documented sources rather than embedded in this report.

The demonstration sources are “Can Open.WAV” by LG (CC BY) and “alarm clock.aif” by hookhead (CC0), processed through ESC-50 (compilation CC BY-NC 3.0), controlled mixing and PCM16 export. The linked execution clip includes sound examples but no author narration; it is a component of the final video, not the completed narrated submission.

AI acknowledgement. I acknowledge using OpenAI Codex (GPT-6 Astra; OpenAI, `https://openai.com/codex/`) and Anthropic Claude (model version not recorded; Anthropic, `https://claude.ai/`). Assistance included substantial code implementation/revision, experiment orchestration, numerical checking, figure/report/slide generation and text drafting; Claude-produced code and review material were incorporated. This extended beyond language editing. I used the lecturer’s feedback to approve the project scope and requested the reported analyses. I remain responsible for the work and its explanation, under the University’s AI acknowledgement guidance.

References

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